

INTRODUCTION

The problem of finding the square root of a graph is a well-studied field in both mathematics and computer-science, with specific relation to distributed computing (e.g., see [PA23]), network design and computational biology. Research in this area has primarily focused around the undirected graph H and its square $G = H^2$ which is obtained by adding an edge between every pair of vertices whose distance in H is at most two. The corresponding decision problem—determining whether a given graph G admits a square root—is NP-complete in general [MS94], but becomes polynomial-time solvable for certain restricted graph classes (e.g., see [LS95]).

BACKGROUND

Let $H = (V, E_H)$ be a finite, simple, undirected graph, where V is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. Denote by $d_H(u, v)$ the *graph distance* defined as the number of edges of the shortest path between vertices u and v in H . If no such path exists, $d_H(u, v) = \infty$.

The *square* of the graph H , denoted by $H^2 = (V, E_{H^2})$, is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \quad (1)$$

That is, two distinct vertices are adjacent in H^2 if and only if they are connected by a path of length one or two in H .

Given a graph $G = (V, E_G)$, if there exists a graph H such that $G = H^2$, then H is called a *square root* of G .

Alternatively, we may approach the problem of studying the square root problem by way of the adjacency matrix of a graph. Let $A_H \in \{0, 1\}^{n \times n}$ denote the adjacency matrix of H , and let $D_H \in \mathbb{R}^{n \times n}$ denote the degree matrix of H . Let $\mathbb{1}_{\geq 1} : \mathbb{R}^{n \times n} \rightarrow \{0, 1\}^{n \times n}$ denote the entrywise *indicator function*, defined by

$$\mathbb{1}_{\geq 1}(M)_{ij} = \begin{cases} 1, & \text{if } M_{ij} \geq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Define the adjacency matrix of G using the adjacency and degree matrices of H

$$A_G = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H). \quad (2)$$

The graph G defined by (2) is equivalent to the graph square H^2 defined in (1), that is,

$$ij \in E_{H^2} \iff 1 \leq d_H(i, j) \leq 2. \quad (3)$$

This matrix formulation naturally leads to an integer programming formulation of the graph square root problem.

CATEGORIZING TREES

In some cases, local graph structure in G can be used to locally reconstruct a root of H . For example, this can be done by analyzing the number of K_3 subgraphs incident to each vertex. For a vertex $s \in V$, define

$$T(s) = \{\{u, v\} \subseteq V \setminus \{s\} \mid (s, u) \in E, (s, v) \in E, (u, v) \in E\}$$

to be the set of K_3 triangles in G that contain s . If $|T(s)| = 1$ where $T(s) = \{\{u, v\}\}$ then the vertex s exists in a unique triangle of G . When the neighbours are structurally distinct with different degree $\deg_G(u) \neq \deg_G(v)$, then the vertex with smaller degree must be the stem vertex in the square root of a graph that is locally acyclic. This can be incorporated in the integer program with the following constraint:

$$x_{sw} = 1, x_{ws} = 1 \text{ where } w = \arg \min\{\deg_G(u), \deg_G(v)\}. \quad (18)$$

Ross and Harary in 1960 gave a polynomial time algorithm for constructing a tree root by using characterizations of the square graph to conclude that existing tree square roots are unique up to isomorphism [RH60].

Proposition 1. *If a square graph G has a tree root H , then the linear program Equations (4)–(17) together with Equation (18) returns an integral solution.*

This proposition gives the first step towards building a unified linear program that can categorize different subclasses of graphs.

FUTURE PLANS

Whilst the algorithmic problem of solving the graph-square problem can be improved drastically by considering heuristics such as rounding or approximation, the primary focus remains on understanding when such heuristics are not required, and when the structure of the graph itself is sufficient to force an integral solution.

The future is to consider further known polynomial subclasses, such as chordal graphs or graphs with girth > 6 , and attempt to find their structural composition such as to generate deterministic constraints which can be added to the linear program.

AIM

Much of the existing research on graph square roots has centered on characterizing when a graph admits such a root and on the computational complexity of reconstructing H from G . In contrast, rather than studying the construction problem directly, we analyze the relationship between the adjacency matrix A_H of the root and the adjacency matrix A_G of the square, and show that the problem can be completely categorized as an integer program. The relaxed linear program can then be solved in polynomial time using interior-point methods. We seek to find subclasses of graphs for which the modified relaxation may yield integral solutions and hence exactly recovers the square root.

INTEGER PROGRAMMING FORMULATION OF GRAPH SQUARE ROOT PROBLEM

Equation (2) permits the construction of an integer program with the objective function $\max \sum_{i=1}^n \sum_{j=1}^n x_{ij}$, variables $x_{ij} \in \mathbb{Z}_{\geq 0}$, $y_{ikj} \in \mathbb{Z}_{\geq 0}$, $z_{ij} \in \mathbb{Z}_{\geq 0}$

for all $i, k, j \in \{1, \dots, n\}$ and constraints

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (4)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (5)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (6)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (7)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (8)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (9)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (10)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (11)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (12)$$

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (13)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (14)$$

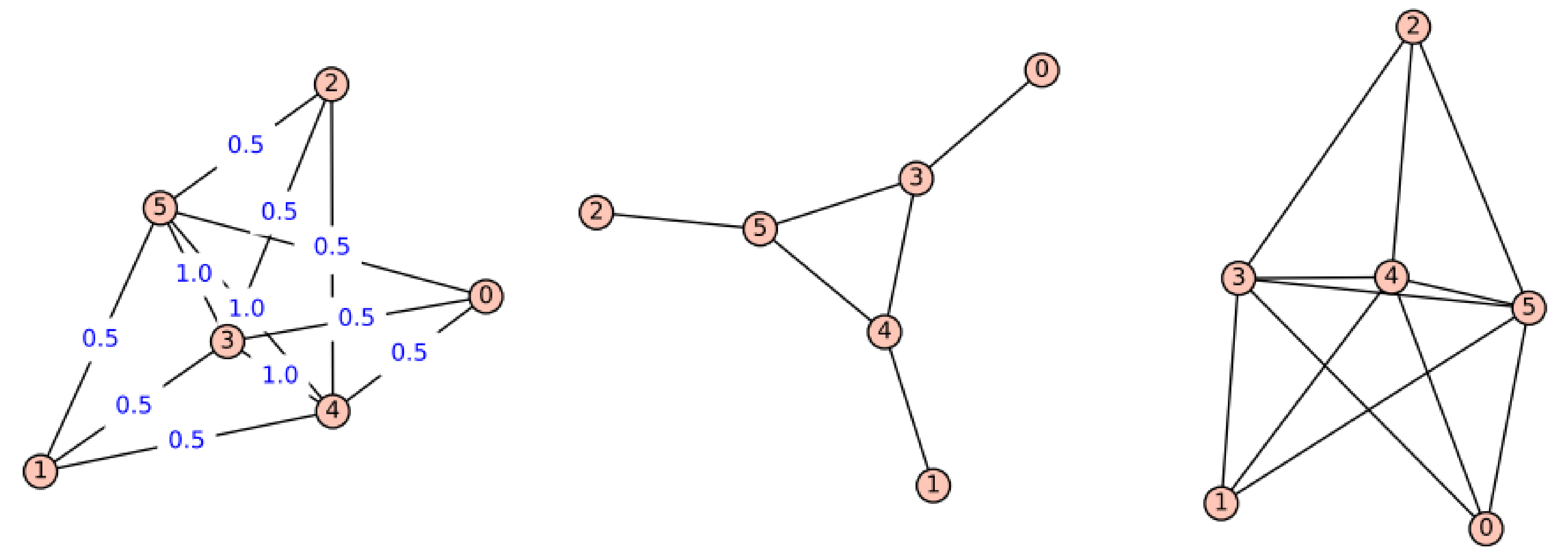
The linear relaxation of the integer program defined by Equations (4)–(14) is constructed in order to generate a feasible weighted solution in polynomial time algorithm. It is done with identical constraints, however the decision variables x_{ij} , y_{ikj} , z_{ij} are replaced with

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (15)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (16)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (17)$$

EXAMPLE



Example of a fractional linear program solution, its true root and the original square.

REFERENCES

- [LS95] Yaw-Ling Lin and Steven S. Skiena. “Algorithms for Square Roots of Graphs”. In: *SIAM Journal on Discrete Mathematics* 8.1 (1995), pp. 99–118.
- [MS94] Rajeev Motwani and Madhu Sudan. “Computing roots of graphs is hard”. In: *Discrete Applied Mathematics* 54.1 (1994), pp. 81–88.
- [PA23] Or Peres and C. Avin. “Distributed Demand-aware Network Design using Bounded Square Root of Graphs”. In: *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications* (2023), pp. 1–10.
- [RH60] Ian C. Ross and Frank Harary. “The square of a tree”. In: *The Bell System Technical Journal* 39.3 (1960), pp. 641–647.